

January 14, 2024

VIA ELECTRONIC SUBMISSION

Division of Dockets Management
Food and Drug Administration
5630 Fishers Lane, Room 1061
Rockville, MD 20852

SCORPION CAPITAL LLC CITIZEN PETITION RE: TRANSMEDICS ORGAN CARE SYSTEM (OCS)

Dear Sir or Madam:

The undersigned, on behalf of Scorpion Capital LLC (“Scorpion” or “Petitioner”), submits this Citizen Petition (“Petition”) pursuant to 21 C.F.R. §10.20 and §10.30 and related relevant provisions of the Federal Food, Drug and Cosmetic Act (the “FDCA”) or the Public Health Service Act or any other statutory provision for which authority has been delegated to the Commissioner of Food and Drugs (“Commissioner”) to request the Commissioner to issue a regulation, request recalls, revise industry guidance, and take such other actions set forth below.

The Organ Care System (OCS) is currently marketed in the U.S. by TransMedics as a normothermic machine perfusion system for the preservation of donor organs, in three versions: OCS Lung, OCS Heart, and OCS Liver. Petitioner recently conducted a six-month-long investigation into TransMedics business practices and the safety/usage of its system, with a particular focus on OCS Liver (PMA P200031). The findings were published on January 10, 2024, as a 342-page investigative report, enclosed as part of this petition as Appendix A and summarized infra.

Petitioner is not the only party troubled by TransMedics practices. As recent background, we reference a letter by Congressman Paul Gosar (R-Arizona) to the company’s CEO, dated Feb 21, 2024, which is publicly available.

Petitioner’s investigation encompassed over 30 research interviews, including: transplant surgeons and hospital administrators, often in leadership roles, at the nation’s pre-eminent and highest-volume transplant centers, such as Massachusetts General Hospital/Harvard, Vanderbilt University, UCSF, University of Chicago, and many others; and former employees of TransMedics. We further reviewed the FDA’s MAUDE database, wherein the paucity of TransMedics submissions is highly suspicious and suggests to us a systematic effort to under-report issues to the FDA; as well as conducting an extensive review of the scientific and clinical literature with regards to normothermic machine perfusion.

Upon this extensive research and investigation, Petitioner believes that:

- 1) The vast majority of OCS Liver usage is off-label, inconsistent with and contraindicated by the label, as well as inconsistent with the pivotal trial which was the basis for FDA Pre-Market Approval. The device is unquestionably not being used in the real-world as it was intended or as studied under the FDA’s approval. Off-label use that directly contradicts the labeled contraindications is particularly alarming, as contraindications are explicitly included to avoid known risks and adverse outcomes. The off-label use appears to be widely-known and acknowledged inside TransMedics, per interviews with ex-employees.
- 2) OCS Liver is prone to safety issues and malfunctions resulting in necrotic livers, which we believe results in injured/damaged livers being transplanted into patients and resulting in poor transplant outcomes. We believe these safety issues, predictably, are the result of and/or exacerbated by the off-label usage.

3) The off-label use is promoted directly by TransMedics, as the company has an unusual, high-risk business model. Its own procurement surgeons retrieve the organ, and then its device technicians manage the organ while it is on the pump. We believe many of these TransMedics procurement surgeons to be unlicensed to practice medicine in the United States, which is itself alarming; and we believe that the device technicians are inexperienced, high-risk, and generally receive only a week of training prior to engaging in what a reasonable person would consider as the practice of medicine. We note that Section 4 of the label ("Warnings and Precautions") states that "Only trained users should use the OCS Liver system." We interviewed several ex-device technicians who indicated that they received insufficient training, and that they resigned because of the company's practices. (p240-248 of our report, Exhibit A).

We furthermore believe that the decision to place livers off-label onto the pump is often, if not almost always, made by TransMedics versus surgeons at recipient centers, who we understand to outsource the decision-making to TransMedics under its end-to-end organ procurement and transportation program. The recipient surgeon is not even in the operating room where organ retrieval occurs, and thus has no ability to examine, palpate, or otherwise determine that a liver meets the labeled criteria and contraindications (such as abnormal anatomy) for OCS machine perfusion. By definition, the decision and therefore the off-label utilization is driven by TransMedics.

4) We believe that TransMedics is engaging in alarming business practices which suggest to us an attempt to conceal safety issues from hospitals and the FDA, as documented in chapter 15 of Petitioner's report.

Petitioner therefore requests that the FDA suspend Pre-Market Approval of TransMedics OCS devices, particularly OCS Liver, which represents the majority of the company's usage and revenue, and undertake additional remedies, as detailed infra (Section A: Action Requested).

Off-label usage (please refer to p155-192, p174-192 in particular, Exhibit A)

- OCS Liver is only indicated for liver allografts which meet three specific criteria for viability: a) from donors ≤ 55 years of age; b) with ≤ 30 minutes of warm ischemic time; and c) with $\leq 15\%$ macrosteatosis.
- Transplant centers which are high-volume users of OCS Liver reported to us that a substantial percentage of livers placed on the device are from donors >55 years of age, typically well into their 60's, and some centers appear to use the device for 70+ year old donors. Several centers stated that 100% of the livers they place on the device are from donors > 55. We note that the average donor age in the pivotal PROTECT OCS Liver trial was 45.9 years, substantially below the donor ages of the livers for which the device is widely used in the real-world.
- High-volume TransMedics OCS Liver users report using the device for livers with warm ischemic times far exceeding 30 minutes. One of its largest users, who stated they do ~100 liver transplants a year after perfusing them on the device, reported ischemic times pushed far beyond the labeled criteria, estimating that 50% of their livers on the device have >30 minutes warm ischemic time; often hitting 45 minutes and as long as two hours. Our report quotes one of the leading liver transplant surgeons in the world, who expressed grave concern if livers exceed 30 minutes of warm ischemic time, given the exponential degradation of the organ beyond that time.
- High-volume TransMedics OCS Liver users report using the device for livers far in excess of the 15% limit for macrosteatosis (fatty liver), pushing it to 50%. Our report cites clinical literature – which is unequivocal on the dangers - indicating that macrosteatosis, along with ischemic time, is a key determinant of transplant outcomes, with risk at even 5% macrosteatosis; substantial levels

of transplant failure at levels >20%; and that livers >30% are simply discarded. One study shows 2-3 fold higher levels of death in patients receiving fatty livers.

- Transplant centers are increasingly using OCS Liver off-label as “back to base,” meaning that livers are retrieved via static cold storage and then placed on the warm perfusion device back at the hospital for scheduling and logistics until surgeons are awake or an operating room becomes available. Surgeons estimate that 1/3 of OCS Liver use is already “back to base.” The device is not indicated for back to base usage; livers must be placed on the device when retrieved, not after an initial, long period of cold storage. Back to base usage was not the basis of the pivotal trial. The consequences are now playing out, with one organ procurement organization reporting to us that 10% of livers in their region are now lost due to damage from this practice, whether via physiologic injury from ice to pump or from excessive perfusion times.
- Ex-employees of TransMedics estimate that 1/3 of livers placed on the pump are specifically contraindicated. Section 3 of the label (“Contraindications”) states that OCS Livers should not be used for split livers or livers with accessory arterial blood supply requiring back table anastomosis, i.e., livers with abnormal anatomy.

Device failures/necrotic livers (please refer to p326-342, Exhibit A)

- Ex-employees, surgeons, and organ procurement organizations indicated to us that: a) TransMedics OCS Liver devices are plagued by failures leading to the loss of a significant percentage of organs; b) that livers in particular are prone to becoming necrotic, essentially rotting on the device, with dead tissue; c) that the issues are prevalent enough that customers question why there hasn’t been an FDA recall; and d) that TransMedics is allegedly engaged in a systematic cover-up, allegedly failing to report device failures to the FDA as required.
- We note that normothermic perfusion-induced liver necrosis is a known risk of the technology, per the clinical literature, as cited in our report.

A. Action Requested

This petition requests that the FDA take the following actions:

1. Suspend FDA Pre-Market Approval of TransMedics OCS devices, the OCS Liver device in particular. The rampant off-label usage indicates that the device is not being used in the manner for which it was intended or studied prior to FDA approval, and therefore must be halted given self-evident risks to transplant patients from livers a) from donors >55 years of age, b) with warm ischemic times >30 minutes, c) macrosteatosis >15%, and d) abnormal anatomy.

2. Re-evaluate the safety and efficacy of the device in light of its actual usage, while current usage is suspended.

a. Convene an Ad Comm to discuss the issues raised in the attached report regarding off-label usage and device failures, and to help guide the agency’s next steps. Petitioner’s investigation encompassed interviews with some of the leading transplant surgeons in the United States, who expressed grave concern regarding these issues, and whose input and guidance would be of benefit to the FDA’s assessment of the issues raised in this Petition.

b. Request data from TransMedics for each liver perfused on its device. TransMedics and each of the 57 Organ Procurement Organizations in the United States, we believe, maintain information for each organ which was placed on machine perfusion, encompassing information such as donor age and other characteristics, as well as serology/pathology results and so forth. While we recognize that the FDA lacks

investigative capabilities to replicate Petitioner's research, it has unique capabilities Petitioner lacks, such as the ability to request data from TransMedics to analyze the extent of off-label usage.

c. Require a new clinical trial to evaluate the safety and efficacy of OCS Liver for livers that reflect its actual usage pattern: >55 years age; >30 minutes warm ischemic time; >15% macrosteatosis; abnormal anatomy. However, we note that such a trial may be reckless and unethical, as livers with these characteristics are high risk and should typically not be transplanted, which is why the label included these criteria in the first place. Thus, we believe the FDA may have no option but to revoke Pre-Market Approval of OCS Liver.

B. Statement Of Grounds

In addition to all of the information detailed supra, and incorporated by reference, Petitioner has enclosed with this Citizen Petition (and incorporates herein) a 342-page investigative report and provides it as its statement of grounds (Exhibit A). The report presents detailed information concerning off-label usage of OCS Liver and its safety issues, in support of the actions requested in Section A.

Background on Petitioner and Petitioner's Report

Scorpion Capital LLC is an investment firm that specializes in investigative due diligence, primarily on biotechnology, pharmaceutical, medical device, and other life sciences companies. Petitioner expends considerable time and resources on each investigation, which typically lasts a minimum of three months and involves interviews with 20-50 experts whom Petitioner identifies and engages. Petitioner believes its investigations have historically been viewed as credible, informative, and helpful by the Department of Justice and the Securities and Exchange Commission.

Petitioner previously submitted a Citizen Petition on Apr 4, 2023, (docket number FDA-2023-P-1273) regarding Harmony Biosciences drug pitolisant, marketed as Wakix. The FDA issued a 13-page final response on June 21, 2024. While Petitioner respectfully disagrees with the FDA's final determination, Petitioner is grateful for the agency's thorough and careful consideration of the issues raised in the Petition, as reflected in its comprehensive response, and hopes it will grant the same assessment of Petitioner's findings regarding TransMedics OCS.

C. Environmental Impact

Petitioner claims a categorical exclusion under 21 C.F.R. § 25.30 and § 25.31, as the relief requested in this Citizen Petition will have no environmental impact and therefore an environmental assessment or environmental impact statement is not required.

D. Economic Impact

Pursuant to 21 C.F.R. § 10.30(b), economic impact information will be submitted by the Petitioner only upon request of the Commissioner following review of this Petition.

E. Certification

The undersigned certifies that, to the best knowledge and belief of the undersigned, this petition includes all the information and views on which the petition relies, and that it includes representative data and information known to the petitioner which are unfavorable to the petition.

Respectfully submitted,

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Appendix A

Petitioner's investigative report dated January 10, 2024. PDF link for public download:

<https://scorpionreports.s3.us-east-2.amazonaws.com/TMDX.pdf>